

## Set — Practice Quiz

Mathematics M.4

Name\_\_\_\_\_

ID\_\_\_\_\_ No.\_\_\_\_\_

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**Answer all questions. Show your work where relevant.**

1. Let  $A = \{1, 2, 3, 4, 5\}$  and  $B = \{3, 4, 5, 6, 7\}$ . Find:
  - 1.1.  $A \cup B$
  - 1.2.  $A \cap B$
  - 1.3.  $A - B$
2. Let  $U = \{1, 2, \dots, 10\}$  be the universal set and  $A = \{2, 4, 6, 8, 10\}$ . Find  $A'$  (the complement of  $A$ ).
3. Write the set  $\{x \mid x \text{ is an even integer, } 1 \leq x \leq 20\}$  in roster form.
4. Let  $A = \{a, b, c\}$ . How many subsets does  $A$  have? List the power set  $P(A)$ .
5. Determine whether each statement is TRUE or FALSE.
  - 5.1.  $\emptyset \subseteq A$  for every set  $A$ .
  - 5.2. If  $A \subseteq B$  and  $B \subseteq A$ , then  $A = B$ .
  - 5.3.  $\{1, 2\} \in \{1, 2, 3\}$
6. In a class of 40 students, 25 study Physics, 18 study Chemistry, and 10 study both. How many students study neither subject?
7. Let  $A$  and  $B$  be finite sets with  $|A| = 12$ ,  $|B| = 9$ , and  $|A \cap B| = 4$ . Find  $|A \cup B|$ .
8. Let  $A = \{1, 2, 3\}$  and  $B = \{x, y\}$ . Find  $A \times B$  (the Cartesian product), and state  $|A \times B|$ .
9. Shade the region representing  $(A \cup B)' \cap C$  on a three-set Venn diagram ( $A, B, C$  all overlapping).
10. If a set has  $n$  elements, how many elements does its power set have? Give a brief reason.

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### Answer Key — Set

1.1.  $A \cup B = \{1,2,3,4,5,6,7\}$    1.2.  $A \cap B = \{3,4,5\}$    1.3.  $A - B = \{1,2\}$

2.  $A' = \{1,3,5,7,9\}$

3.  $\{2,4,6,8,10,12,14,16,18,20\}$

4. 8 subsets:  $\emptyset$ ,  $\{a\}$ ,  $\{b\}$ ,  $\{c\}$ ,  $\{a,b\}$ ,  $\{a,c\}$ ,  $\{b,c\}$ ,  $\{a,b,c\}$

5.1. True   5.2. True   5.3. False

6.  $40 - (25 + 18 - 10) = 40 - 33 = 7$  students

7.  $12 + 9 - 4 = 17$

8.  $A \times B = \{(1,x),(1,y),(2,x),(2,y),(3,x),(3,y)\}$ ,  $|A \times B| = 6$

9. (Diagram — the part of C outside both A and B)

10.  $2^n$ , since each element is either included or not, independently.